

Forecasting Techniques

Those who have knowledge don't predict. Those Who Predict Don't have Knowledge.

—Lao Tzu

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- Understand the importance of forecasting and its impact on the effectiveness of the supply chain and overall performance of an organization.
- Learn various components of time-series data such as trend, seasonality, cyclical component, and random component.
- Learn different techniques such as moving average, exponential smoothing, and Croston's method.
- Learn Auto-Regression (AR), Moving Average (MA), and Auto-Regressive Integrated Moving Average models (ARIMA).
- Learn about the practical challenges associated with forecasting models using case studies.

Importance of Forecasting

Forecasting is one of the most important and frequently addressed problems in analytics. Inaccurate forecasting can have significant impact on both top line and bottom line of an organization. For example, non-availability of a product in the market can result in customer

dissatisfaction, whereas, too much inventory can erode the organization's profit. Thus, it becomes necessary to forecast the demand for a product and service as accurately as possible.

13.1 | Introduction to Forecasting

Forecasting is by far the most important and frequently used application of predictive analytics since it has significant impact on both top line and bottom line of an organization. Every organization prepares long-range and short-range planning for the organization and forecasting demand for product and service is an important input for both long-range and short-range planning. Different capacity planning problems such as manpower planning, machine capacity, warehouse capacity, Materials Requirement Planning (MRP) will depend on the forecasted demand for the product/service. Budget allocation for marketing promotions and

advertisement are usually made based on forecasted demand for the product. Forecasting can be very challenging due to several factors that influence the demand and scale of business with Stock Keeping Units (SKUs) running into several millions. For example:

1. Boeing 747-400 has more than six million parts and several thousand unique parts (Hill, 2011). Forecasting demand for spare parts is important since non-availability of mission critical parts can result in Aircraft on Ground (AOG) which can be very expensive for airlines.
2. Amazon.com sells more than 350 million products through its E-commerce portal. Amazon itself sells about 13 million SKUs and has more (about two million) retailers selling their products through its platform (Ali, 2017). Predicting demand for these products is important since overstocking can impact the bottom line and under stocking can result in customer dissatisfaction. Amazon.com may not stock all SKUs they sell through their portal since most are sold by their suppliers (online marketplace) directly to the customers, but even if they have to predict demand for products directly sold by them, the number of SKUs is 13 million (during 2017).
3. Walmart sells more than 142,000 products through their supercenters (source: Walmart website¹). Being a brick-and-mortar retail store, Walmart does not have the advantages of Amazon.com (being also a market place, Amazon do not have to predict demand for all products sold through its portal). They have to maintain stock for each and every product sold by Walmart and predict demand for the products as accurately as possible.
4. Demand for products and service is not the only application of forecasting. Even manpower planning requires the use of sophisticated models. Indian information technology (IT) companies struggle to manage the right level of manpower for each skill required to manage their business. This would involve forecasting business opportunities, skills required to manage current and future projects, and so on.
5. Many products may have intermittent demands, that is, the gap between two demands can be long and the gap itself may be random. The modeler has to forecast the next instance of demand and the actual demand quantity when demand occurs, making it much more difficult to forecast.
6. One of the innovative applications of forecasting was the Netflix forecasting contest in which the participants were challenged to forecast the movie rating (on a scale of 1 to 5) likely to be given by a viewer for a movie. An accurate customer movie rating forecast can further be used for movie recommendations to customers.

13.2 | Time-Series Data and Components of Time-Series Data

Time-series data is a data on a response variable, Y_t , such as demand for a spare parts of a capital equipment or a product or a service or market share of a brand observed at different time points t . Whenever we have a forecasting problem, we will use time-series data. The variable Y_t is a random variable. The data points or measurements are usually collected at regular intervals and are arranged in chronological order. If the time-series data contains observations of just a single variable (such as demand of a product at time t), then it is termed univariate time series. If the data consists of more than one variable, for example, demand for a product at time t , price at time t , amount of money spent by the company on promotion at time t , competitors price at time t , etc. then it is called multivariate time-series data.

From a forecasting perspective, time-series data can be broken into the following components [Figures 13.1(a)–(d)]:

1. **Trend Component (T_t):** Trend is the consistent long-term upward or downward movement of the data over a period of time.
2. **Seasonal Component (S_t):** Seasonal component (measured using seasonality index) is the repetitive upward or downward movement (or fluctuations) from the trend that

¹ Source: http://corporate.walmart.com/_news_/news-archive/2005/01/07/our-retail-divisions

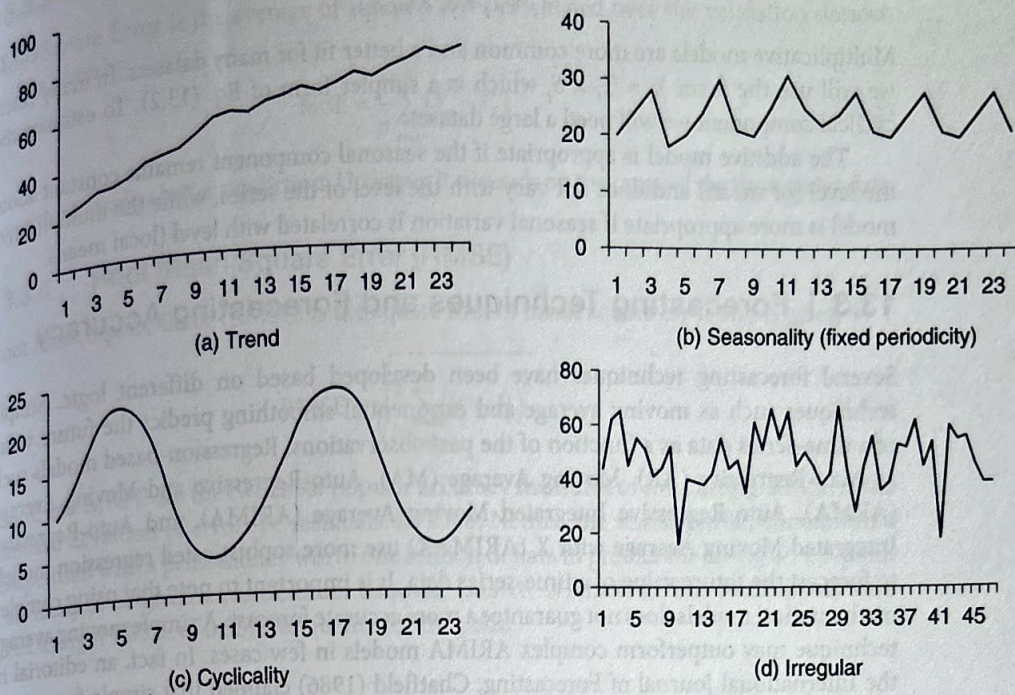


Figure 13.1 | Components in time-series data.

occurs within a calendar year such as seasons, quarters, months, days of the week, etc. The upward or downward fluctuation may be caused due to festivals, customs within a society, school holidays, business practices within the market such as 'end of season sale', and so on. For example, in India the demand for many products surge during the festival months of October and November. A similar pattern exists during December in many countries due to Christmas. Usually, for a given context seasonal fluctuation occurs at fixed intervals (such as months, quarters) known as periodicity of seasonal variation and repeats over time.

3. **Cyclical Component (C_t)**: Cyclical component is fluctuation around the trend line that happens due to macro-economic changes such as recession, unemployment, etc. Cyclical fluctuations have repetitive patterns, with a time lag between repetitions of more than a year. In the case of seasonality, fluctuations are observed within a calendar year and are driven by factors such as festivals and customs that exist in a society. A major difference between seasonal and cyclical fluctuation is that the former occurs at a fixed period within a calendar year, whereas the latter is driven by random intervals between fluctuations. That is, the periodicity of seasonal fluctuations is constant, while the periodicity of cyclical fluctuations is not constant.
4. **Irregular Component (I_t)**: Irregular component is the white noise or random uncorrelated changes that follow a normal distribution with mean value of 0 and constant variance.

The time-series data can be modelled as an addition of the above components or product of the above components. The additive time-series model is given by

$$Y_t = T_t + S_t + C_t + I_t \quad (13.1)$$

Additive models assume that seasonal and cyclical components are independent of the trend component. Additive models are not very common since in many cases the seasonal component may not be independent of trend. At the Indian Institute of Management Bangalore (IIMB) there are many weekend programs and the number of students enrolled in these programs is fixed. The demand for food at the canteens of IIMB increases by a fixed quantity on Saturdays. This increase is additive in nature.

The multiplicative time-series model is given by

$$Y_t = T_t \times S_t \times C_t \times I_t \quad (13.2)$$

Multiplicative models are more common and a better fit for many datasets. In many cases, we will use the form $Y_t = T_t \times S_t$, which is a simpler form of Eq. (13.2). To estimate the cyclical component we will need a large dataset.

The additive model is appropriate if the seasonal component remains constant about the level (or mean) and does not vary with the level of the series, while the multiplicative model is more appropriate if seasonal variation is correlated with level (local mean).

13.3 | Forecasting Techniques and Forecasting Accuracy

Several forecasting techniques have been developed based on different logic. Simple techniques such as moving average and exponential smoothing predict the future value of a time-series data as a function of the past observations. Regression-based models such as Auto-Regressive (AR), Moving Average (MA), Auto-Regressive and Moving Average (ARMA), Auto-Regressive Integrated Moving Average (ARIMA), and Auto-Regressive Integrated Moving Average with X (ARIMAX) use more sophisticated regression models to forecast the future value of a time-series data. It is important to note that using complex mathematical models does not guarantee a more accurate forecast. A simple moving average technique may outperform complex ARIMA models in few cases. In fact, an editorial in the International Journal of Forecasting, Chatfield (1986) claimed that simple forecasting models sometimes outperform complex models.

Usually, many different forecasting techniques such as moving average, exponential smoothing, and ARIMA are used for forecasting before selecting the best model. Model selection may depend on the chosen forecasting accuracy measure. The following four forecasting accuracy measures are frequently used:

1. Mean absolute error
2. Mean absolute percentage error
3. Mean squared error
4. Root mean square error

In the following subsections, we will discuss these measures.

13.3.1 Mean Absolute Error (MAE)

Mean Absolute Error (MAE) is the average absolute error and should be calculated on the validation dataset. Assume that the validation data has n observations and forecasting is carried out on these n observations using the model developed. The mean absolute error is given by

$$MAE = \frac{\sum_{t=1}^n |Y_t - F_t|}{n} \quad (13.3)$$

In Eq. (13.3), Y_t is the actual value of Y at time t and F_t is the corresponding forecasted value.

13.3.2 Mean Absolute Percentage Error (MAPE)

Mean Absolute Percentage Error (MAPE) is the average of absolute percentage error. Assume that the validation data has n observations and the forecasting is carried out on these n observations. The mean absolute percentage error is given by

$$MAPE = \frac{1}{n} \sum_{t=1}^n \frac{|Y_t - F_t|}{Y_t} \times 100\% \quad (13.4)$$

MAPE defined in Eq. (13.4) is one of the popular forecasting accuracy measures used by practitioners since it expresses the average error in percentage terms and is easy to interpret. Since MAPE is dimensionless it can be used to compare models with varying scales.

13.3.3 Mean Square Error (MSE)

Mean Square Error is the average of squared error calculated over the validation dataset. MSE is given by

$$\text{MSE} = \frac{1}{n} \sum_{t=1}^n (Y_t - F_t)^2 \quad (13.5)$$

Lower MSE implies better prediction. However, it depends on the range of the time-series data.

13.3.4 Root Mean Square Error (RMSE)

Root Mean Square Error (RMSE) is the square root of mean square error and is given by

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (Y_t - F_t)^2} \quad (13.6)$$

RMSE and MAPE are the two most popular accuracy measures of forecasting. RMSE is the standard deviation of errors or residuals. In 2006, Netflix, the movie portal, announced a competition with a prize money worth one million dollars to predict the rating on a 5-point scale likely to be given a customer for a movie² (source: Wikipedia). The participants were given a target RMSE of 0.8572 to qualify for the prize.

13.4 | Moving Average Method

Moving average is one of the simplest forecasting techniques which forecasts the future value of a time-series data using average (or weighted average) of the past 'N' observations. Mathematically, a simple moving average is calculated using the formula

$$F_{t+1} = \frac{1}{N} \sum_{k=t+1-N}^t Y_k \quad (13.7)$$

The above formula is called Simple Moving Average (SMA) since 'N' past observations are given equal weights (1/N). In a weighted moving average, past observations are given differential weights (usually the weight decrease as the data becomes older). Weighted moving average is given by

$$F_{t+1} = \sum_{k=t+1-N}^t W_k \times Y_k \quad (13.8)$$

where W_k is the weight given to value of Y at time k (Y_k) and $\sum_{k=t+1-N}^t W_k = 1$.

We Sell Beauty (WSB) is a manufacturer and distributor of health and beauty products. WSB is interested in forecasting demand for 'Kesh', their shampoo brand sold in 100 ml bottles. WSB believes that the monthly demand for 'Kesh' depends on the promotion expenditure (in thousands of rupees) and whether the competition was on promotion or not during that month. The data for 48 months (starting from January 2012) is shown in Table 13.1. The table has the quantity of 100 ml bottles sold during the month, promotion expenses (in thousands of rupees) incurred by the company, and whether the competition was on promotion (value of 1 implies that the competition was on promotion and 0 otherwise). Use Simple Moving Average with $N = 12$ and forecast the demand of Kesh for months 37 to 48. Calculate the values of MAPE and RMSE.

Example 13.1

² https://en.wikipedia.org/wiki/Netflix_Prize

Table 13.1 | Data on sales of shampoo, promotion expenses (in 1000 of rupees), and dummy variable for promotion by competition

Month	Sale Quantity	Promotion Expenses	Competition Promotion	Month	Sale Quantity	Promotion Expenses	Competition Promotion
1	3002666	105	1	25	4634047	165	0
2	4401553	145	0	26	3772879	129	1
3	3205279	118	1	27	3187110	120	1
4	4245349	130	0	28	3093683	112	1
5	3001940	98	1	29	4557363	162	0
6	4377766	156	0	30	3816956	140	1
7	2798343	98	1	31	4410887	160	0
8	4303668	144	0	32	3694713	139	0
9	2958185	112	1	33	3822669	141	1
10	3623386	120	0	34	3689286	136	0
11	3279115	125	0	35	3728654	130	1
12	2843766	102	1	36	4732677	168	0
13	4447581	160	0	37	3216483	121	1
14	3675305	130	0	38	3453239	128	0
15	3477156	130	0	39	5431651	170	0
16	3720794	140	0	40	4241851	160	0
17	3834086	167	1	41	3909887	151	1
18	3888913	148	1	42	3216438	120	1
19	3871342	150	1	43	4222005	152	0
20	3679862	129	0	44	3621034	125	0
21	3358242	120	0	45	5162201	170	0
22	3361488	122	0	46	4627177	160	0
23	3670362	135	0	47	4623945	168	0
24	3123966	110	1	48	4599368	166	0

Moving average forecast for the period $n = 37$ to 48 is given by

$$F_{t+1} = \frac{1}{12} \sum_{k=t+1-12}^t Y_k, \text{ for } t = 36, 37, \dots, 47$$

The forecasted values using 12 period moving average and the corresponding RMSE and MAPE calculations are given in Table 13.2.

Table 13.2 | Simple moving average forecast, RMSE, and MAPE calculations

Month	Actual Demand (Y_t)	Forecasted Demand (F_t)	$(Y_t - F_t)^2$	$ Y_t - F_t / F_t$
37	3216483	3928410	5.07E + 11	0.221337
38	3453239	3810280	1.27E + 11	0.103393
39	5431651	3783643	2.72E + 12	0.303408
40	4241851	3970688	7.35E + 10	0.063926
41	3909887	4066369	2.45E + 10	0.040022
42	3216438	4012413	6.34E + 11	0.247471
43	4222005	3962370	6.74E + 10	0.061496

(Continued)

Table 13.2 | (Continued)

Month	Actual Demand (Y_t)	Forecasted Demand (F_t)	$(Y_t - F_t)^2$	$ Y_t - F_t / F_t$
44	3621034	3946629	1.06E + 11	0.089918
45	5162201	3940490	1.49E + 12	0.236665
46	4627177	4052117	3.31E + 11	0.124279
47	4623945	4130275	2.44E + 11	0.106764
48	4599368	4204882	1.56E + 11	0.08577

The RMSE using the moving average forecast is given by 734725.8359 and the MAPE value is 0.1403 (or 14.03%). The graph of actual and forecasted demand is shown in Figure 13.2.

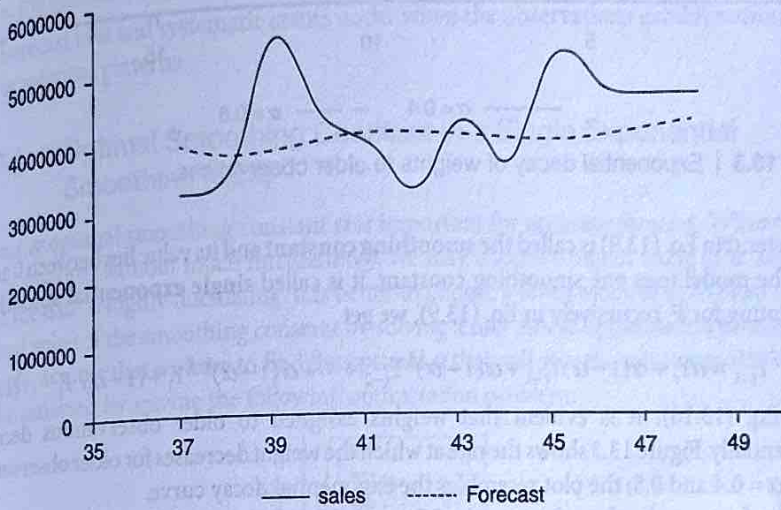


Figure 13.2 | Plot of actual sales forecasted sales using moving average.

In moving average an important decision that a one has to take is the number of periods, N . Forecast accuracy will depend on the chosen N . If N is small, the average tends to be more sensitive to recent observations or more responsive to recent trend. So, if responsiveness is important, then relatively few data points may be included. This would enable the moving average to make adjustments with the changes in the data quickly, though at times it would also be responding to just the random noise in the data. On the other hand, if N is large, that is more data points are included, the forecast will be less sensitive or responsive to recent changes in the data. Since the moving average will always be centered around the range of data points considered, it will lag behind the trend until about $(N + 1)/2$ time periods.

13.5 | Single Exponential Smoothing (SES)

One of the drawbacks of Simple Moving Average technique is that it gives equal weight to all previous observations used in forecasting future value. This can be overcome by assigning differential weights to the past observations [Eq. (13.8)]. One easier way to assign differential weight is achieved by using Single Exponential Smoothing (SES) technique (also known as simple exponential smoothing). Just like the moving average, SES assumes a fairly steady time-series data with no significant trend, seasonal or cyclical component. Here, the weights assigned to past data decline exponentially with the most recent observations assigned higher weights.

In single ES, the forecast at time $(t + 1)$ is given by (Winters, 1960)

$$F_{t+1} = \alpha Y_t + (1 - \alpha) F_t \quad (13.9)$$

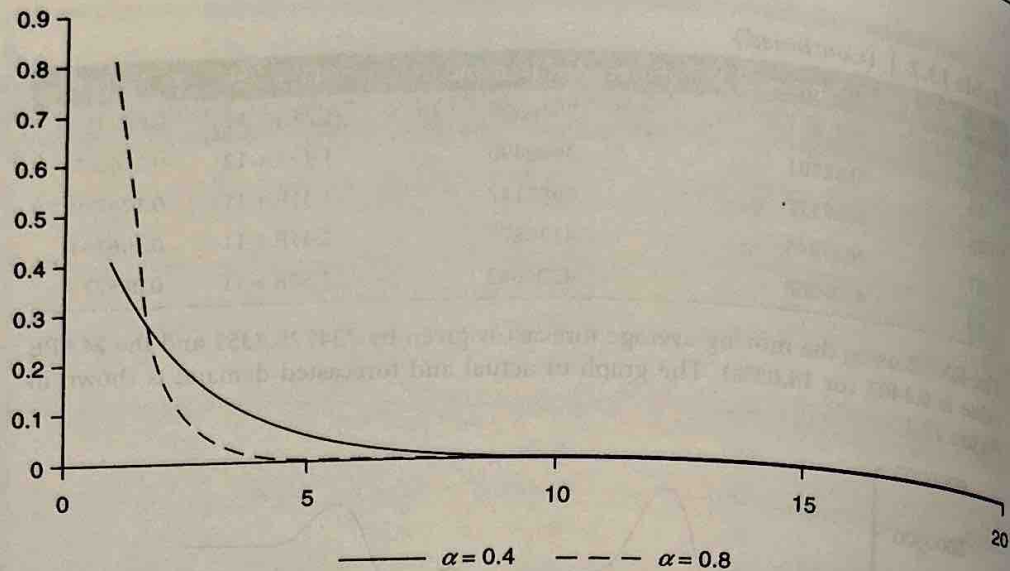


Figure 13.3 | Exponential decay of weights to older observations.

Parameter α in Eq. (13.9) is called the **smoothing constant** and its value lies between 0 and 1. Since the model uses one smoothing constant, it is called **single exponential smoothing**. Substituting for F_t recursively in Eq. (13.9), we get

$$F_{t+1} = \alpha Y_t + \alpha(1-\alpha)Y_{t-1} + \alpha(1-\alpha)^2 Y_{t-2} + \dots + \alpha(1-\alpha)^{t-1} Y_1 + (1-\alpha)^t F_1 \quad (13.10)$$

From Eq. (13.10), it is evident that weights assigned to older observations decrease exponentially. Figure 13.3 shows the rate at which the weight decreases for older observations when $\alpha = 0.4$ and 0.8 ; the plot resembles the exponential decay curve.

The forecasted values for months 37 to 48 for the data in Table 13.1 using simple exponential smoothing is shown in Table 13.3. Exponential smoothing uses the entire historical data. To begin exponential smoothing we will need the forecast for the F_t in Eq. (13.9). We can use $F_t = Y_t$ or use moving average to forecast the initial forecast F_t . The forecasted value for period 2 is given by

$$F_2 = \alpha Y_1 + (1-\alpha) F_1$$

Table 13.3 | Forecast for periods 37 to 48 using single exponential smoothing ($\alpha = 0.2$)

Month	Actual Demand	Forecasted Demand $\alpha = 0.2$	$(Y_t - F_t)^2$	$\frac{ Y_t - F_t }{Y_t}$
37	3216483	3980905	5.8434E + 11	0.237658
38	3453239	3828020	1.4046E + 11	0.10853
39	5431651	3753064	2.8177E + 12	0.309038
40	4241851	4088781	2.343E + 10	0.036086
41	3909887	4119395	4.3894E + 10	0.053584
42	3216438	4077494	7.4142E + 11	0.267705
43	4222005	3905283	1.0031E + 11	0.075017
44	3621034	3968627	1.2082E + 11	0.095993
45	5162201	3899108	1.5954E + 12	0.244681
46	4627177	4151727	2.2605E + 11	0.102752
47	4623945	4246817	1.4223E + 11	0.08156
48	4599368	4322243	7.6799E + 10	0.060253

We will assume F_1 same as Y_1 . Thus the value of F_2 will be same as Y_1 , that is 3002666. The forecasted values using single exponential smoothing with $\alpha = 0.2$ are shown in Table 13.3. The RMSE using the single exponential smoothing with $\alpha = 0.2$ is given by 742339.222 and the MAPE value is 0.1394 (or 13.94%).

In summary, single exponential smoothing technique has the following advantages:

1. It uses all historic data unlike the moving average where only the past few observations are considered to predict future value.
2. It assigns progressively decreasing weights to older data.

Some disadvantages of smoothing methods are:

1. Increasing n makes forecast less sensitive to changes in data.
2. It always lags behind trend as it is based on past observations. The longer the time period n , the greater the lag as it is slow to recognize the shifts in the level of the data points.
3. Forecast bias and systematic errors occur when the observations exhibit a strong trend or seasonal patterns.

13.5.1 Optimal Smoothing Constant in a Single Exponential Smoothing (SES)

Choosing optimal smoothing constant α is important for accurate forecast. Whenever the data is smooth (without much fluctuations), we may choose a higher value of α . However, when the data is highly fluctuating, it is better to choose a lower value of α . We can find the optimal value of the smoothing constant by solving a non-linear optimization problem. For example, assume that we have to find the optimal α that will give the minimum RMSE. This can be achieved by solving the following optimization problem:

$$\text{Min}_{\alpha} \left[\sqrt{\frac{1}{n} \sum_t (Y_t - F_t)^2} \right] \quad (13.11)$$

subject to the constraint: $0 < \alpha < 1$. For the data in Table 13.1, the optimal value of α that minimizes the RMSE is 0.1574 and the corresponding RMSE is 739399.76. Table 13.4 shows the forecasted value, RMSE, and MAPE calculations for $\alpha = 0.1574$ (rounded to 4 decimals).

Table 13.4 | Forecast for periods 37 to 48 using single exponential smoothing ($\alpha = 0.1574$)

Month	Actual Demand	Forecasted Demand $\alpha = 0.1574$	$(Y_t - F_t)^2$	$\frac{ Y_t - F_t }{Y_t}$
37	3216483	3931892	5.1181E + 11	0.22242
38	3453239	3819242	1.3396E + 11	0.105988
39	5431651	3761610	2.789E + 12	0.307465
40	4241851	4024580	4.7207E + 10	0.051221
41	3909887	4058792	2.2173E + 10	0.038084
42	3216438	4035345	6.7061E + 11	0.254601
43	4222005	3906397	9.9608E + 10	0.074753
44	3621034	3956094	1.1227E + 11	0.092532
45	5162201	3903334	1.5847E + 12	0.243862
46	4627177	4101560	2.7627E + 11	0.113594
47	4623945	4184325	1.9327E + 11	0.095075
48	4599368	4253549	1.1959E + 11	0.075188

13.6 | Double Exponential Smoothing – Holt's Method

One of the drawbacks of single exponential smoothing is that the model does not do well in the presence of trend. This can be improved by introducing an additional equation to capture the trend in time-series data. Double exponential smoothing uses two equations to forecast the future values of the time series, one for forecasting the level (short term average value) and another for capturing the trend. The two equations are provided in Eqs. (13.12) and (13.13).

Level (or Intercept) equation (L_t):

$$L_t = \alpha \times Y_t + (1 - \alpha) \times F_t \quad (13.12)$$

The trend equation is given by (T_t)

$$T_t = \beta \times (L_t - L_{t-1}) + (1 - \beta) \times T_{t-1} \quad (13.13)$$

α and β are the smoothing constants for level and trend, respectively, and $0 < \alpha < 1$ and $0 < \beta < 1$.

For the data in Table 13.1, the forecast at time $t + 1$ is given by

$$F_{t+1} = L_t + T_t \quad (13.14)$$

$$F_{t+n} = L_t + nT_t \quad (13.15)$$

where L_t is the level which represents the smoothed value up to and including the last data, T_t is the slope of the line or the rate of increase or decrease at period t , n is the number of time periods into the future.

The initial of L_t is usually taken same the as Y_t (that is, $L_t = Y_t$). The starting value of T_t can be taken as $(Y_t - Y_{t-1})$ or the difference between two previous actual values of observations prior to the period for which forecasting is carried out. Another option for T_t is $(Y_t - Y_1)/(t - 1)$.

The value of

$$L_1 = Y_1 = 3002666$$

and

$$T_1 = (Y_{36} - Y_1)/35 = (4732677 - 3002666)/35 = 49428.8857$$

The value of

$$F_2 = L_1 + T_1 = 3002666 + 49428.8857 = 3052095$$

The forecasted values for periods 37 to 48 are shown in Table 13.5 ($\alpha = 0.0328$ and $\beta = 0.9486$).

The RMSE and MAPE of the forecast using double exponential smoothing is given by 659888.9554 and 0.1135 (11.35%). The values of α and β used in Table 13.5 are optimized values of α and β that minimize the root mean square error.

Table 13.5 | Forecasted values using double exponential smoothing ($\alpha = 0.0328$ and $\beta = 0.9486$)

Month	Actual Demand	L_t	T_t	$F_t (= L_{t-1} + T_{t-1})$	$(Y_t - F_t)^2$	$\frac{ Y_t - F_t }{Y_t}$
37	3216483	3678293	66894.6916	3693955	2.27979E+11	0.148445
38	3453239	3735612	57810.9617	3745188	85234318285	0.084544
39	5431651	3847157	108782.913	3793423	2.68379E+12	0.301608
40	4241851	3965318	117678.771	3955940	81745109031	0.067402
41	3909887	4077319	112292.624	4082997	29966946139	0.044275
42	3216438	4157691	82013.2329	4189611	9.47066E+11	0.302562
43	4222005	4239124	81462.532	4239704	313269245.9	0.004192
44	3621034	4297641	59696.6025	4320586	4.89374E+11	0.193191
45	5162201	4383737	84739.1839	4357338	6.47805E+11	0.155915
46	4627177	4473682	89677.0074	4468476	25185883916	0.034298
47	4623945	4565346	91562.092	4563359	3670690475	0.013103
48	4599368	4655021	89771.7849	4656908	3310862728	0.01251

13.7 | Triple Exponential Smoothing (Holt-Winter Model)

Moving average and single and double exponential smoothing techniques discussed so far can handle data as long as it does not have any seasonal component associated with it. However, when there is seasonality in the time-series data, techniques such as moving average, exponential smoothing, and double exponential smoothing are no longer appropriate. In most cases, the fitted error values (actual demand minus forecast) associated with simple exponential smoothing and Holt's method will indicate systematic error patterns that reflect the existence of seasonality. For example, the presence of seasonality may result in all positive errors, except for negative values that occur at fixed intervals. Such pattern in error would imply the existence of seasonality. Such time series data require the use of a seasonal method to eliminate the systematic patterns in error.

Triple exponential smoothing is used when the data has trend as well as seasonality. The following three equations which account for level, trend, and seasonality are used for forecasting (for a multiplicative model, Winters 1960):

Level (or Intercept) equation:

$$L_t = \alpha \frac{Y_t}{S_{t-c}} + (1 - \alpha) [L_{t-1} + T_{t-1}] \quad (13.16)$$

Trend equation:

$$T_t = \beta \times (L_t - L_{t-1}) + (1 - \beta) T_{t-1} \quad (13.17)$$

Seasonal equation:

$$S_t = \gamma \frac{Y_t}{L_t} + (1 - \gamma) S_{t-c} \quad (13.18)$$

The forecast F_{t+1} using triple exponential smoothing is given by

$$F_{t+1} = [L_t + T_t] \times S_{t+1-c} \quad (13.19)$$

where c is the number of seasons (if it is monthly seasonality, then $c = 12$; in case of quarterly seasonality $c = 4$; and in case of daily data $c = 7$). The initial values of L_t and T_t are calculated using the following equations:

$$L_t = Y_t \quad (13.20)$$

Alternatively

$$L_t = \frac{1}{c} (Y_1 + Y_2 + \dots + Y_c) \quad (13.21)$$

$$T_t = \frac{1}{c} \left[\frac{Y_t - Y_{t-c}}{12} + \frac{Y_{t-1} - Y_{t-c-1}}{12} + \frac{Y_{t-2} - Y_{t-c-2}}{12} + \dots + \frac{Y_{t-c+1} - Y_{t-2c+1}}{12} \right] \quad (13.22)$$

Several techniques exist to calculate the initial seasonality index (Winters, 1961; Makridakis *et al.*, 1998; Taylor 2011). The initial seasonality index can be calculated using a technique called method of simple averages which is described in Section 13.7.1. Several variations to the procedure discussed in Section 13.7.1 exist, such as ratio-to-moving average.

13.7.1 Predicting Seasonality Index Using Method of Averages

The following steps are used to predict the seasonality index using method of averages:

Step 1

Calculate the average of value of Y for each season (that is, if the data is monthly, then we need to calculate the average for each month based on the training data). Let these averages be $\bar{Y}_1, \bar{Y}_2, \dots, \bar{Y}_c$.

Step 2

Calculate the average of the seasons' averages calculated in Step 1 (say $\bar{\bar{Y}}$).

Step 3

The seasonality index for season k is given by the ratio $\bar{Y}_k / \bar{\bar{Y}}$. Variation to the procedure explained above is to first divide the value of Y_t with its yearly average and calculate the seasonal average. We will use the first three years of data in Table 13.1 to calculate the seasonality index for various months. The seasonality index based on first 3 years of data using method of averages is shown in Table 13.6.

Table 13.6 | Seasonality index using method of averages

Month	Sale Quantity (2012)	Sale Quantity (2013)	Sale Quantity (2014)	Monthly Average $\bar{Y}_k$	Seasonality Index $\frac{\bar{Y}_k}{\bar{\bar{Y}}}$
January	3002666	4447581	4634047	4028098.00	1.087932
February	4401553	3675305	3772879	3949912.33	1.066815
March	3205279	3477156	3187110	3289848.33	0.888541
April	4245349	3720794	3093683	3686608.67	0.9957
May	3001940	3834086	4557363	3797796.33	1.02573
June	4377766	3888913	3816956	4027878.33	1.087872
July	2798343	3871342	4410887	3693524.00	0.997568
August	4303668	3679862	3694713	3892747.67	1.051375
September	2958185	3358242	3822669	3379698.67	0.912808
October	3623386	3361486	3689286	3558053.33	0.960979
November	3279115	3670362	3728654	3559377.00	0.961337
December	2843766	3123966	4732677	3566803.00	0.963342
Average of monthly averages				3702528.22	

Seasonality index can be interpreted as percentage change from the trend line. For example, the seasonality index for January is approximately 1.088 or 108.8%. This implies that in January, the demand will be approximately 8.8% more from the trend line. The seasonality index for March is 0.8885 or 88.85%. This implies that the demand in March will be 11.15% less from the trend line. Note that, the multiplicative model is used in this example.

To start the triple exponential smoothing, we need to set the starting values of level and trend.

$$L_{36} = Y_{36} / S_{36} = 4732677 / 0.9633 = 4912983.494$$

The initial value of trend (T_{36}) can be calculated based on the second and third years by using Eq. (13.22):

$$T_{36} = \frac{1}{12} \left[\frac{Y_{36} - Y_{24}}{12} + \frac{Y_{35} - Y_{23}}{12} + \frac{Y_{34} - Y_{22}}{12} + \dots + \frac{Y_{25} - Y_{13}}{12} \right]$$

$$T_{36} = \frac{1}{12} \left[\frac{4732677 - 3123966}{12} + \frac{3728654 - 3670362}{12} + \dots + \frac{4634047 - 4447581}{12} \right] = 21054.35$$

The forecast for Period 37 using triple exponential smoothing is given by

$$F_{37} = [L_{36} + T_{36}] \times S_{37-12} = [L_{36} + T_{36}] \times S_{25} \quad (13.23)$$

The seasonal index S_{25} (seasonality index for January) is 1.088. Substituting the values of L_{36} , T_{36} and S_{25} , we get

$$F_{37} = [4912983.494 + 21054.35] \times 1.088 = 5368233.2$$

Table 13.7 | Forecasting using triple exponential smoothing (values differ for different round off values of parameters)

Month t	Actual Demand	L_{t-1}	T_{t-1}	S_t	F_t	$(Y_t - F_t)^2$	$\frac{ Y_t - F_t }{Y_t}$
37	3216482	4912983.49	21054.35	1.09	5367895.97	4.62858E+12	0.668872
38	3453239	4301229.28	-295349.93	1.07	4273531.48	6.7288E+11	0.237543
39	5431651	3759825.78	-418376.71	0.89	2969014.38	6.06458E+12	0.453386
40	4241851	4228345.39	25071.45	1.00	4235127.90	45200134.5	0.001585
41	3909887	4255577.53	26151.79	1.03	4391900.21	2.32337E+11	0.123281
42	3216437	4131354.31	-49035.71	1.09	4441041.44	1.49966E+12	0.380733
43	4222004	3722098.55	-229145.74	1.00	3484457.63	5.43975E+11	0.174691
44	3621034	3729543.06	-110850.61	1.05	3804603.81	33697874118	0.050695
45	5162201	3562820.55	-138786.56	0.91	3125486.18	4.14821E+12	0.394544
46	4627176	4138038.05	218215.47	0.96	4186269.09	1.94399E+11	0.095286
47	4623945	4503072.73	291625.07	0.96	4609319.06	213918263.4	0.003163
48	4599368	4799566.34	294059.34	0.96	4906905.01	94579010434	0.066865

The forecast for the period 37 to 48 for the data in Table 13.1 is given in Table 13.7. Note that the values such as seasonality index are rounded to two decimals. Forecast values will be different if the actual seasonality index values in Table 13.6 are used.

The RMSE and MAPE using triple exponential smoothing are 1228588.29 and 0.2208 (22.08%), respectively. The values of $\alpha = 0.32$, $\beta = 0.5$, and $\gamma = 1$ are used to calculate the level, trend, and seasonal components. It is important to note that exponential smoothing techniques are very sensitive to initial values of level, trend, and seasonal index.

13.8 | Croston's Forecasting Method for Intermittent Demand

Products such as spare parts may have intermittent demands. Exponential smoothing models discussed so far in the chapter will produce biased estimate when used for intermittent demand. Croston (1972) developed a model that uses two separate exponential smoothing equations to predict the mean time between demands and the magnitude of demand whenever the demand occurs. That is, Croston's method has two components: (a) Predicting time between demand and (b) magnitude of the demand. The primary objective of Croston's method is to forecast mean demand per period. Let

Y_t = Demand at time t (Y_t may take value 0)

F_t = Forecasted demand

TD_t = Time between the latest and the previous non-zero demand in period t

FTD_t = Forecasted time between demand at period t

The following steps are used for forecasting demand:

$$\text{If } Y_t = 0 \text{ then } F_{t+1} = F_t \text{ and } FTD_{t+1} = FTD_t \quad (13.24)$$

$$\text{If } Y_t \neq 0 \text{ then } F_{t+1} = \alpha \times Y_t + (1 - \alpha)F_t \text{ and } FTD_{t+1} = \beta \times TD_t + (1 - \beta) \times FTD_t \quad (13.25)$$

α and β are smoothing constants for forecasted demand and forecasted time between demands, respectively. Once the forecasted demand and time between demands are known, then the mean demand per period, D_{t+1} , is given by

$$D_{t+1} = \frac{F_{t+1}}{FTD_{t+1}} \quad (13.26)$$

Example 13.2

Quarterly demand for spare parts of avionics system of an aircraft is shown in Table 13.8. Use the demand during the quarters 1 to 4 as training data to forecast demand for periods 5 to 16 using Croston's method.

Table 13.8 | Quarterly demand for avionic system spares

Quarter	1	2	3	4	5	6	7	8
Demand	20	12	0	18	16	0	20	22
Quarter	9	10	11	12	13	14	15	16
Demand	0	28	0	0	30	26	0	34

Procedure used for starting values of F_t and FTD_t is shown in Table 13.9.

Table 13.9 | Calculating initial values in Croston's method

Quarter	Demand	TD_t	FTD_t	F_t
1	20			
2	12	1		
3	0			
4	18	2	1.5	16.67

In Table 13.9, $TD_4 = 2$ since the elapsed time from the previous demand to current demand period is 2 ($4 - 2$). The forecasted time between demand is the average TD_t values up to $t = 4$. So, $FTD_4 = (1+2)/2 = 1.5$. The forecasted demand F_4 for $t = 4$ is $(20 + 12 + 18)/3 = 16.67$. Note that the total value is divided by three (not four) since only three quarters had non-zero demand. So, the starting values for Croston's method are

$$TD_4 = 2, FTD_4 = 1.5, \text{ and } F_4 = 16.67$$

Let $\alpha = \beta = 0.2$. Then

$$F_5 = 0.2 \times 18 + (1 - 0.2) \times 16.67 = 16.936$$

$$FTD_5 = 0.2 \times 2 + (1 - 0.2) \times 1.5 = 1.6$$

The forecasted values for the remaining quarters are shown in Table 13.10.

Table 13.10 | Forecasted demand for periods 5 to 16 using Croston's method

Quarter	Demand	TD_t	FTD_t	F_t	$D_t = (F_t / FTD_t)$
1	20				
2	12	1			
3	0				
4	18	2	1.5000	16.67	11.11333
5	16	1	1.6000	16.936	10.585
6	0		1.4800	16.7488	11.31676
7	20	2	1.4800	16.7488	11.31676
8	22	1	1.5840	17.39904	10.98424
9	0		1.4672	18.31923	12.48585
10	28	2	1.4672	18.31923	12.48585
11	0		1.5738	20.25539	12.8707
12	0		1.5738	20.25539	12.8707
13	30	3	1.5738	20.25539	12.8707
14	26	1	1.8590	22.20431	11.94417
15	0		1.6872	22.96345	13.61034
16	34	2	1.6872	22.96345	13.61034

13.9 | Regression Model for Forecasting

Regression is probably a more appropriate method for forecasting when the data has values of predictor (explanatory) variables in addition to the dependent variable Y_t . In the data provided in Table 13.1, we also have information such as promotion expenses and whether the competition was on promotion or not. Using the values of these predictor variables is likely to give a better forecast than the exponential smoothing techniques discussed in previous sections. Parker and Segura (1971) claimed that regression method can predict more accurately than less scientific methods such as exponential smoothing. The forecasted value at time t , F_t , can be written as a regression equation as follows:

$$F_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \dots + \beta_n X_{nt} + \epsilon_t \quad (13.27)$$

Here F_t is the forecasted value of Y_t and X_{1t} , X_{2t} , etc. are the predictor variables measured at time t . The regression equation for Example 13.1 is

$$F_t = \beta_0 + \beta_1 \text{promotion expenses at time } t + \beta_2 \text{competition promotion at time } t$$

For the data in Table 13.1, the regression outputs using SPSS are shown in Tables 13.11 and 13.12. The model is developed based on the first 36 months data.

Table 13.11 | Model summary

Model	R	R-Square	Adjusted R-Square	Std. Error of the Estimate	Durbin-Watson
1	0.928	0.862	0.853	207017.359	1.608

The R-square for the model is 0.862 (note that we will need high R-square value for forecasting applications) and the Durbin-Watson statistic value is 1.608. Since this is a time-series data we need to check whether the errors, ϵ_t , are correlated (auto-correlation). For $n = 36$ (sample size) and number of predictor variables = 2, the Durbin-Watson critical values are $d_L = 1.153$ and $d_U = 1.376$. Since the model Durbin-Watson statistic $D = 1.608$ ($4 - D = 2.392$) lies within d_U and $(4 - d_U)$, we can conclude that there is no auto-correlation. Whenever regression model is used for forecasting, it should be checked for auto-correlation among the errors. Presence of auto-correlation may lead to inclusion of a non-significant variable in the model since the standard error of the regression coefficient is underestimated when auto-correlation of errors is present.

Table 13.12 | Coefficients

Model	Unstandardized Coefficients		Standardized Coefficients		t	Sig.
	B	Std. Error	Beta			
(Constant)	808471.843	278944.970			2.898	0.007
1 Promotion Expenses	22432.941	1953.674	0.825		11.482	0.000
Competition Promotion	-212646.036	77012.289	-0.198		-2.761	0.009

The regression model (based on values in Table 13.12) is given by

$$F_t = 808471.843 + 22432.941X_{1t} - 212646.036X_{2t} \quad (13.28)$$

where

$$X_{1t} = \text{Promotion expenses at time } t$$

$$X_{2t} = \begin{cases} 1 & \text{Competition is on promotion} \\ 0 & \text{Otherwise} \end{cases}$$

As expected, sales increases as promotion expenses increase and decreases whenever the competition is on promotion. The forecasted values for period 37 to 48 using the regression The RMSE and MAPE based on regression model are 302969 and 0.0419 (or 4.19%),

Table 13.13 | Forecasts using regression model

Period	Y_t	X_{1t}	X_{2t}	F_t	$(Y_t - F_t)^2$	$\frac{ Y_t - F_t }{Y_t}$
37	3216483	121	1	3310211.67	8785063205	
38	3453239	128	0	3679888.29	5.137E+10	0.02914
39	5431651	170	0	4622071.81	6.5542E+11	0.065634
40	4241851	160	0	4397742.4	2.4302E+10	0.149048
41	3909887	151	1	3983199.9	5374781013	0.036751
42	3216438	120	1	3287778.73	5089499329	0.018751
43	4222005	152	0	4218278.88	13884007.5	0.02218
44	3621034	125	0	3612589.47	71310120.7	0.000883
45	5162201	170	0	4622071.81	2.9174E+11	0.002332
46	4627177	160	0	4397742.4	5.264E+10	0.104632
47	4623945	168	0	4577205.93	2184540571	0.049584
48	4599368	166	0	4532340.05	4492746215	0.010108
						0.014573

respectively. The RMSE and MAPE for regression based forecasting are much smaller than the values we obtained so far using moving average and exponential smoothing techniques. For Example 13.1, the moving average method resulted in an RMSE of 734725.84 and MAPE is 14.03%. The RMSE and MAPE for single exponential smoothing are 742339.22 and 13.94%, respectively. The plot of actual demand and forecasted demand using regression model is shown in Figure 13.4.

13.9.1 Forecasting Time-Series Data with Seasonal Variation

As mentioned earlier, one can expect seasonal variation in demand for many products and services. The following steps are used to forecast time-series data with seasonal variations:

Step 1

Estimate the seasonality index (using techniques such as method of averages).

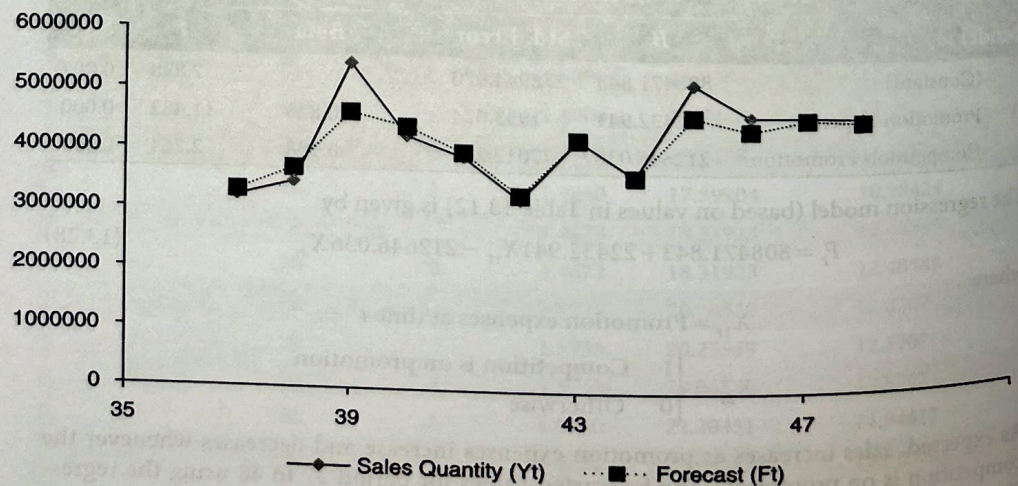


Figure 13.4 | Actual sales quantity and forecasted sales using regression model.

Step 2

De-seasonalize the data using either additive or multiplicative model. For example, in multiplicative model, the de-seasonalized data $Y_{d,t} = Y_t / S_t$, where $Y_{d,t}$ is the de-seasonalized data and S_t is the seasonality index for period t .

Step 3

Develop a forecasting model on the de-seasonalized data ($F_{d,t}$).

Step 4

The forecast for period $t + 1$ is $F_{t+1} = F_{d,t+1} \times S_{t+1}$.

Hiccup Viking (HV) is the Vice President of Viking Cookies that specializes in chocolate chip cookies (Choco-Chip). Viking Cookies believes that demand for cookies is seasonal and is driven by several factors such as school holidays, festivals, etc. The shelf life of Choco-Chip cookies is six months and excess inventory and running out of stock can have financial impact. Hiccup would like to develop a forecasting model that can be used to forecast demand. The past monthly demand (quantity of 200 gram packets) for four years (January 2013 to December 2016) along with average price per unit during that month is shown in Table 13.14. Develop a forecasting model using regression to predict demand between months 37 and 48, given that the data is seasonal.

Example 13.3

Table 13.14 Monthly demand (quantity of 200 gram packets) along with average price per unit

Period	Month	Demand in Units	Average Price	Period	Demand in Units	Average Price
1	January	10500472	37	25	10658309	36
2	February	10123572	34	26	8677622	38
3	March	7372141	36	27	7330354	37
4	April	7764303	38	28	8115471	37
5	May	6904463	40	29	8481936	34
6	June	10068862	34	30	8778999	37
7	July	6436190	40	31	10145039	32
8	August	9898436	34	32	8497839	38
9	September	6803825	39	33	8792138	34
10	October	8333787	36	34	8485358	36
11	November	7541964	39	35	8575904	36
12	December	8540662	37	36	9885156	32
13	January	10229437	37	37	11023467	35
14	February	8453201	38	38	7942451	40
15	March	7997459	35	39	12492798	32
16	April	8557825	35	40	9756258	32
17	May	7818397	36	41	8992741	32
18	June	8944499	37	42	7397807	40
19	July	8904086	36	43	9710611	32
20	August	8463682	39	44	8328379	39
21	September	7723957	37	45	11873063	32
22	October	7731422	39	46	10642507	32
23	November	8441834	35	47	10635075	32
24	December	7485122	40	48	10578547	32

Solution

Since the demand is seasonal, the first step in forecasting is to estimate the seasonality index. We can use data from the first 36 months to estimate the seasonality index using method of averages explained in Section 13.7.1. Table 13.15 gives the seasonality index for various months. For example, the seasonality index for January is 1.2251. That is, in January the demand will increase by 22.51% from the trend.

Table 13.15 | Seasonality index for various months

Month	Demand (2012)	Demand (2013)	Demand (2014)	Average	Seasonality Index
1	10500472	10229437	10658309	10462739	1.2251
2	10123572	8453201	8677622	9084798	1.0637
3	7372141	7997459	7330354	7566651	0.8860
4	7764303	8557825	8115471	8145866	0.9538
5	6904463	7818397	8481936	7734932	0.9057
6	10068862	8944499	8778999	9264120	1.0847
7	6436190	8904086	10145039	8495105	0.9947
8	9898436	8463682	8497839	8953319	1.0483
9	6803825	7723957	8792138	7773307	0.9102
10	8333787	7731422	8485358	8183522	0.9582
11	7541964	8441834	8575904	8186567	0.9585
12	8540662	7485122	9885156	8636980	1.0113
Average of monthly averages				8540659	

De-seasonalized data is calculated by dividing the value of Y_t with the corresponding seasonality index. The de-seasonalized data for periods 1 to 48 is shown in Table 13.16.

Table 13.16 | De-seasonalized demand (seasonality index from Table 13.15 is rounded to 2 decimals)

Month	Demand	Seasonality Index	De-seasonalized Demand	Month	Demand	Seasonality Index	De-seasonalized Demand
1	10500472	1.23	8571459.88	25	10658309	1.23	8700301.09
2	10123572	1.06	9517214.68	26	8677622	1.06	8157870.71
3	7372141	0.89	8321110.54	27	7330354	0.89	8273944.56
4	7764303	0.95	8140603.02	28	8115471	0.95	8508790.52
5	6904463	0.91	7623682.26	29	8481936	0.91	9365476.36
6	10068862	1.08	9282556.42	30	8778999	1.08	8093422.43
7	6436190	0.99	6470703.29	31	10145039	0.99	10199440.54
8	9898436	1.05	9442215.37	32	8497839	1.05	8106172.12
9	6803825	0.91	7475473.63	33	8792138	0.91	9660065.60
10	8333787	0.96	8697481.33	34	8485358	0.96	8855667.03
11	7541964	0.96	7868174.77	35	8575904	0.96	8946835.52
12	8540662	1.01	8445415.13	36	9885156	1.01	9774915.11
13	10229437	1.23	8350215.95	37	11023467	1.23	8998376.94
14	8453201	1.06	7946891.53	38	7942451	1.06	7466733.21
15	7997459	0.89	9026921.81	39	12492798	0.89	14100917.65

(Continued)

Table 13.16 | (Continued)

Month	Demand	Seasonality Index	De-seasonalized Demand	Month	Demand	Seasonality Index	De-seasonalized Demand
16	8557825	0.95	8972583.38	40	9756258	0.95	10229098.91
17	7818397	0.91	8632818.30	41	8992741	0.91	9929490.54
18	8944499	1.08	8245998.07	42	7397807	1.08	6820091.57
19	8904086	0.99	8951833.08	43	9710611	0.99	9762682.98
20	8463682	1.05	8073589.43	44	8328379	1.05	7944522.57
21	7723957	0.91	8486437.69	45	11873063	0.91	13045128.20
22	7731422	0.96	8068828.55	46	10642507	0.96	11106956.05
23	8441834	0.96	8806966.63	47	10635075	0.96	11095071.36
24	7485122	1.01	7401646.68	48	10578547	1.01	10460573.30

Regression output for the de-seasonalized demand and average price using Microsoft Excel are shown in Table 13.17.

Table 13.17 | Regression output using SPSS for data in Table 13.16 (based on first 36 cases)

Model	Unstandardized Coefficients			
	B	Std. Error	T	Sig.
1 (Constant)	20812014.673	717702.417	28.998	0.000
Average Price	-335945.859	19616.915	-17.125	0.000

Regression model for demand forecasting based on the first 36 months of de-seasonalized data is given by

$$F_{d,t} = 20812014.673 - 335945.859 \times \text{Average Price}$$

The forecasted values are given in Table 13.18.

Table 13.18 | Forecasted values for the data in Table 13.15

Month	Demand	Seasonality Index (S_t)	De-seasonalized Demand	$F_{d,t}$	$F_t = F_{d,t} \times S_t$	$(Y_t - F_t)^2$	$\frac{ Y_t - F_t }{Y_t}$
37	11023467	1.2251	8998377	9053910	11091497	4628131462	0.006171
38	7942451	1.0637	7466733	7374180	7844001	9692313467	0.012395
39	12492798	0.8860	14100918	10061747	8914269	1.2806×10^{13}	0.286447
40	9756258	0.9538	10229099	10061747	9596642	2.5477×10^{10}	0.01636
41	8992741	0.9057	9929491	10061747	9112521	1.4347×10^{10}	0.01332
42	7397807	1.0847	6820092	7374180	7998831	3.6123×10^{11}	0.081244
43	9710611	0.9947	9762683	10061747	10008080	8.8488×10^{10}	0.030633
44	8328379	1.0483	7944523	7710126	8082657	6.0379×10^{10}	0.029504
45	11873063	0.9102	13045128	10061747	9157730	7.373×10^{12}	0.228697
46	10642507	0.9582	11106956	10061747	9641005	1.003×10^{12}	0.094104
47	10635075	0.9585	11095071	10061747	9644592	9.8106×10^{11}	0.093134
48	10578547	1.0113	10460573	10061747	10175223	1.6267×10^{11}	0.038127

RMSE and MAPE values are 1381119.09 and 0.0775 (7.75%), respectively.

13.10 | Auto-Regressive (AR), Moving Average (MA) and ARMA Models

Auto-Regressive (AR) and Moving Average (MA) models are popular models that are frequently used for forecasting. AR and MA models are combined to create models such as Auto-Regressive Moving Average (ARMA) and Auto-Regressive Integrated Moving Average (ARIMA) models. The initial ARMA and ARIMA models were developed by Box and Jenkins in 1970 (Box and Jenkins, 1970). ARMA models are basically regression models; auto-regression means regression of a variable on itself measured at different time periods. One of the fundamental assumptions of the AR model is that the time series is assumed to be a stationary process. If a time-series data, Y_t , is stationary, then it satisfies the following conditions:

1. The mean values of Y_t at different values of t are constant.
2. The variances of Y_t at different time periods are constant (Homoscedasticity).
3. The covariances of Y_t and Y_{t-k} for different lags depend only on k and not on time t .

When the time series data is not stationary (that is, any one of the above conditions are not satisfied), non-stationary times-series data must be converted to stationary data before applying AR models. Another important concept associated with forecasting with regression-based models is the white noise of residuals. White noise is a process of residuals ε_t that are uncorrelated and follow normal distribution with mean 0 and constant standard deviation. In AR models, one of the important assumptions we make is that errors follow a white noise.

13.11 | Auto-Regressive (AR) Models

Auto-regression is regression of a variable on itself measured at different time points. Auto-Regressive Model with lag 1, AR(1), is given by

$$Y_{t+1} = \beta Y_t + \varepsilon_{t+1} \quad (13.29)$$

which can be re-written as

$$Y_{t+1} - \mu = \beta \times (Y_t - \mu) + \varepsilon_{t+1} \quad (13.30)$$

where ε_{t+1} is a sequence of uncorrelated residuals that follow normal distribution with zero mean and constant standard deviation. $Y_{t+1} - \mu$ can be interpreted as deviation from mean value μ and is known as mean centered series. Equation (13.30) can be expanded recursively as shown in Eqs. (13.31) and (13.32):

$$Y_{t+1} - \mu = \beta \times [\beta \times (Y_{t-1} - \mu) + \varepsilon_t] + \varepsilon_{t+1} \quad (13.31)$$

$$Y_{t+1} - \mu = \beta^t (Y_0 - \mu) + \beta^{t-1} \varepsilon_1 + \beta^{t-2} \varepsilon_2 + \dots + \beta \varepsilon_t + \varepsilon_{t+1} \quad (13.32)$$

Equation (13.32) can be written as

$$Y_{t+1} - \mu = \beta^t (Y_0 - \mu) + \sum_{k=1}^{t-1} \beta^{t-k} \times \varepsilon_k + \varepsilon_{t+1} \quad (13.33)$$

In Eq. (13.33), if $|\beta| > 1$ in the first part on the right-hand side $[\beta^t (Y_0 - \mu)]$ will result in an infinitely large value of Y_{t+1} as the value of t increases and is not very useful for practical applications. The value of $|\beta| = 1$ would imply that the future value of Y depends on the entire past (and will lead to non-stationarity). For practical applications, the value of $|\beta|$

should be less than one. The second part of Eq. (13.33), $\sum_{k=1}^{t-1} \beta^{t-k} \times \varepsilon_k$, can also become infinitely large if the errors do not follow a white noise. When the errors are white noise

then the expected value of $\sum_{k=1}^{t-1} \beta^{t-k} \times \varepsilon_k$ is zero. The coefficient β in Eq. (13.30) can be estimated using ordinary least squares estimate. The sum of squared errors is given by

$$\sum_{t=2}^n \varepsilon_t^2 = \sum_{t=2}^n [(Y_t - \mu) - \beta \times (Y_{t-1} - \mu)]^2 \quad (13.34)$$

Taking first-derivative of Eq. (13.34) with respect to β and equating that to zero, the estimate of β is given by

$$\hat{\beta} = \frac{\sum_{t=2}^n (Y_t - \mu)(Y_{t-1} - \mu)}{\sum_{t=2}^n (Y_{t-1} - \mu)^2} \quad (13.35)$$

We can generalize an auto-regressive model with 1 lag, AR(1), to Auto-Regressive model with p lags, AR(p) process is given by

$$Y_{t+1} = \beta_0 + \beta_1 Y_t + \beta_2 Y_{t-1} + \dots + \beta_p Y_{t-p+1} + \varepsilon_{t+1} \quad (13.36)$$

The forecasted value is given by

$$F_{t+1} = \hat{\beta}_0 + \hat{\beta}_1 Y_t + \hat{\beta}_2 Y_{t-1} + \dots + \hat{\beta}_p Y_{t-p+1} \quad (13.37)$$

where $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots$ are the estimated values $\beta_0, \beta_1, \beta_2, \dots$ and so on. Note that software such as SPSS use the mean centered series [that is, equation of the form (13.30)] while estimating the parameters and hence forecasting the future values should be carried out using Eq. (13.30).

13.11.1 AR Model Identification: ACF and PACF

One of the important tasks in using Auto-Regressive model in forecasting is model identification, which is, identifying the value of p (the number of lags). Among the standard approaches used for model identification is Auto-Correlation Function (ACF) and Partial Auto-Correlation Function (PACF). Auto-correlation is the correlation between Y_t measured at different time periods (for example, Y_t and Y_{t-1} or Y_t and Y_{t-k}). Auto-correlation indicates the memory of a process, that is, how far in time it can remember what has happened before. The auto-correlation of k -lags (correlation between Y_t and Y_{t-k}) is given by

$$\rho_k = \frac{\sum_{t=k+1}^n (Y_{t-k} - \bar{Y})(Y_t - \bar{Y})}{\sum_{t=1}^n (Y_t - \bar{Y})^2} \quad (13.38)$$

where n is the number of observations in the sample. A plot of auto-correlation for different values of k is called **Auto-Correlation Function (ACF)** or **correlogram**.

Partial Auto-Correlation of lag k (ρ_{pk}) is the correlation between Y_t and Y_{t-k} when the influence of all intermediate values ($Y_{t-1}, Y_{t-2}, \dots, Y_{t-k+1}$) is removed (partial out) from both Y_t and Y_{t-k} . A plot of partial auto-correlation for different values of k is called **Partial Auto-Correlation Function (PACF)**. Hypothesis tests can be carried out to check whether the auto-correlation and partial auto-correlation values are different from zero. The corresponding null and alternative hypotheses are

$H_0: \rho_k = 0$ and $H_A: \rho_k \neq 0$, where ρ_k is the auto-correlation of order k

$H_0: \rho_{pk} = 0$ and $H_A: \rho_{pk} \neq 0$, where ρ_{pk} is the partial auto-correlation of order k

The null hypothesis is rejected when $|\rho_k| > 1.96/\sqrt{n}$ and $|\rho_{pk}| > 1.96/\sqrt{n}$. To select the appropriate p in the auto-regressive model, the following thumb rule may be used. The number of lags is p when (Yaffee and McGee, 2000).

1. The partial auto-correlation, $|\rho_{pk}| > 1.96/\sqrt{n}$ for first p values (first p lags) and cuts off to zero.
2. The auto-correlation function (ACF), ρ_k , decreases exponentially.

Note that model identification is an iterative process and may require additional inputs. Model identification using ACF and PACF cannot be taken as conclusive evidence for the number of lags in the AR process.

Example 13.4

Dr Dawai Sundari (DS) is concerned about the amount of food wasted at her Die Another Day (DAD) hospital. DAD hospital prepares food for all patients, which accounted for 4% of the operating cost. Dr DS found that as high as 50% of the food prepared was wasted on a few days. She believed an accurate forecasting model would help her reduce food wastage. The demand for continental breakfast at DAD hospital for past 37 days is shown in Table 13.19. Build an auto-regressive model based on the first 30 days of data and forecast the demand for continental breakfast from days 31 to 37. Comment on the accuracy of the forecast.

Table 13.19 | Demand for continental breakfast at DAD

Day	Demand CB	Day	Demand CB
1	25	20	43
2	25	21	41
3	25	22	46
4	35	23	41
5	41	24	40
6	30	25	32
7	40	26	41
8	40	27	41
9	40	28	40
10	40	29	43
11	40	30	46
12	40	31	45
13	44	32	45
14	49	33	46
15	50	34	43
16	45	35	40
17	40	36	41
18	42	37	41
19	40		

Solution

The first step in AR model building is identification of the right value of p using ACF and PACF plots. ACF and PACF based on the first 30 observations are given in Figures 13.5 and 13.6, respectively. The horizontal lines in the plot represent the upper and lower critical values for ρ_k and ρ_{pk} . The correlation values (vertical bars) beyond the critical values will result in rejection of the null hypothesis.

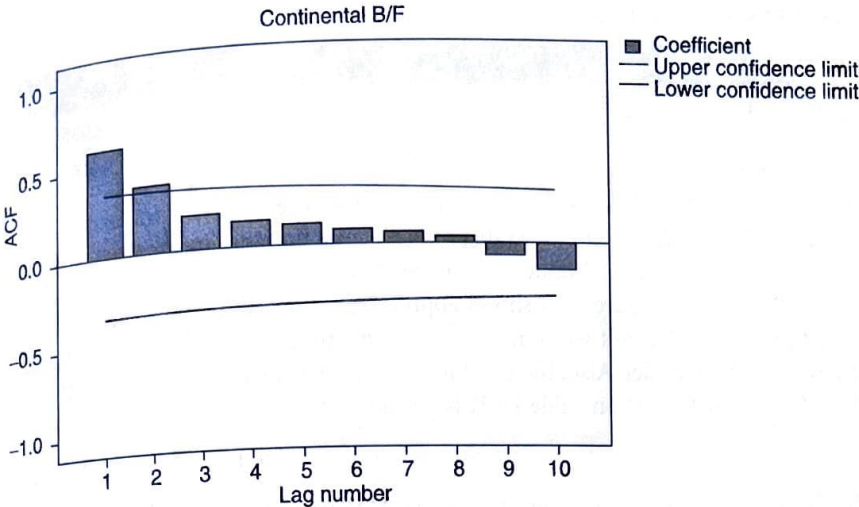


Figure 13.5 | ACF plot for demand for continental breakfast.

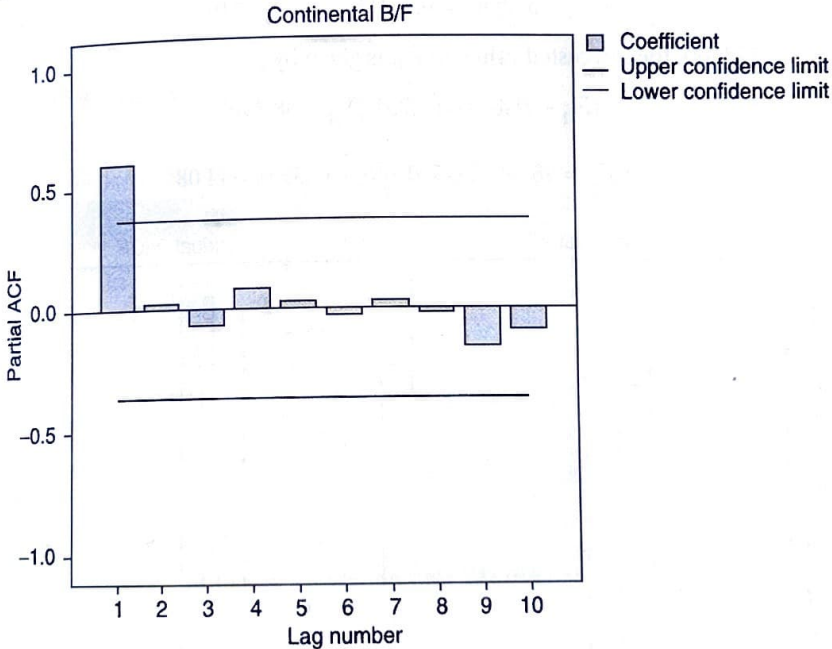


Figure 13.6 | PACF plot for demand for continental breakfast.

In Figure 13.6, the PACF values cut-off to zero after lag 1 and in Figure 13.5 ACF, the values of auto-correlations decrease exponentially. Thus, we can conclude that the value of p in this case is 1 (note that this is a thumb rule, the correct model may be different from what we identify using ACF and PACF plots). The presence of outliers may have a serious impact on ACF and PACF and thus the identification of lag value (value of p). The values of R^2 , RMSE, MAPE, and regression parameter estimates of AR(1), using SPSS are shown in Tables 13.20 and 13.21.

Table 13.20 | AR(1) model statistics

Model	Model Fit Statistics			
	R-Square	RMSE	MAPE	Normalized BIC
Continental B/F-Model_1	0.373	5.133	10.518	3.498

Table 13.21 | ARIMA model parameters

			Estimate	SE	T	Sig.
Continental	Continental	Constant	38.890			
B/F-Model_1	B/F	AR		2.995	12.985	0.000
		Lag 1	0.731	0.130	5.616	0.000

The residual ACF and PACF plots are shown in Figure 13.7. It is important that the residual ACF and PACF follow a white noise. Since all the auto- (and partial) correlation values are within the critical values, we can conclude that residuals are uncorrelated. The normal P-P plot in Figure 13.8 shows approximate normal distribution for residuals. Since the residual follows white noise as evident from Figures 13.7 and 13.8, we can accept the AR(1) model. Also, the p -value for lag 1 term in Table 13.21 is less than 0.05. The AR(1) model based on Table 13.21 is given by

$$(F_{t+1} - 38.890) = 0.731(Y_t - 38.890) \quad (13.39)$$

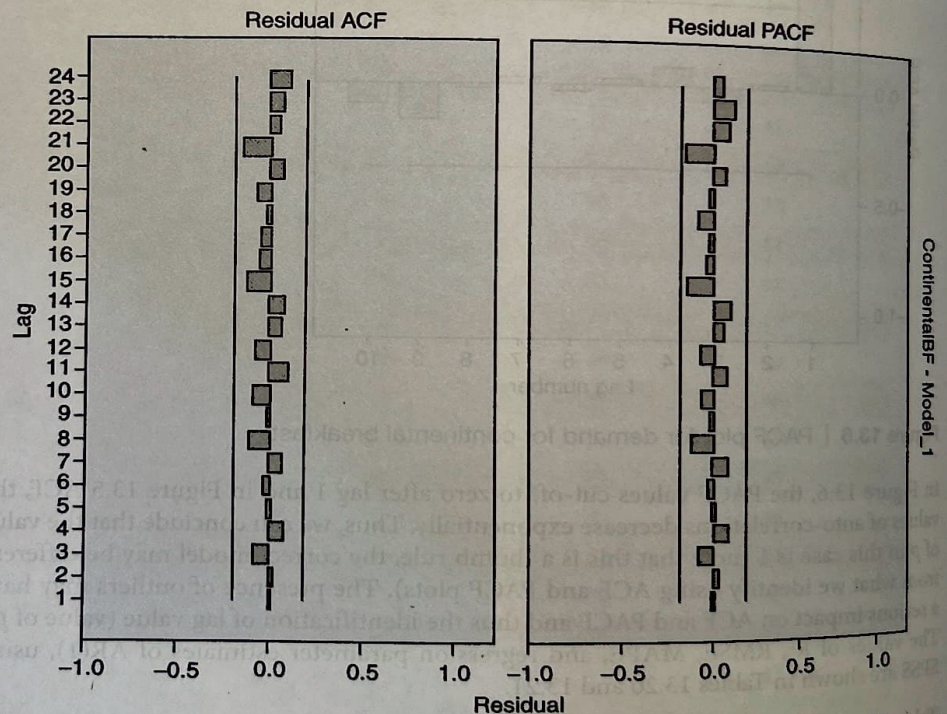
The following equation can be used when we need to forecast several periods in future using forecasted values repetitively.

$$(F_{t+k} - 38.890) = 0.731(F_{t+k-1} - 38.890) \quad (13.40)$$

Using Eq. (13.39), the forecasted value for F_{31} is given by

$$(F_{31} - 38.890) = 0.731(Y_{30} - 38.890)$$

$$\Rightarrow F_{31} = 38.890 + 0.731(46 - 38.890) = 44.08$$

**Figure 13.7 | ACF and PACF plot of residuals.**

The forecasted values for periods 31–37 and RMSE and MAPE calculations are shown in Table 13.22.

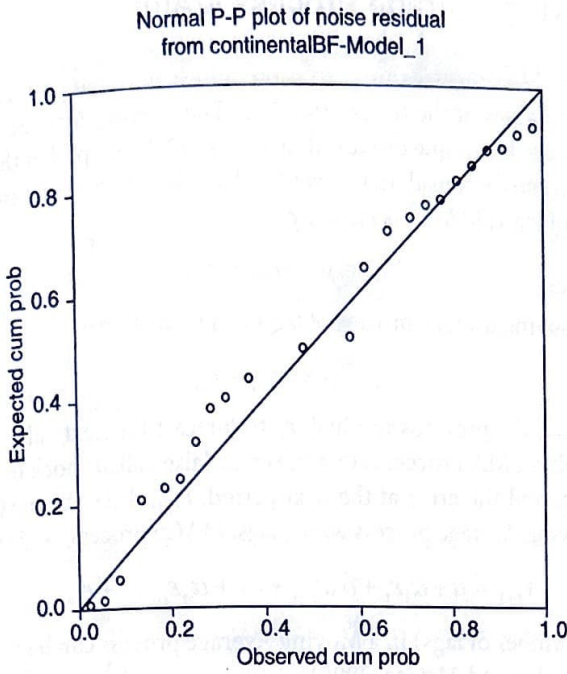


Figure 13.8 | Normal P-P plot of residuals.

Table 13.22 | AR(1) model forecast

Day	Y_t	F_t	$(Y_t - F_t)^2$	$ Y_t - F_t /Y_t$
31	45	44.08741	0.832821	0.02028
32	45	43.35641	2.701388	0.036524
33	46	43.35641	6.988568	0.057469
34	43	44.08741	1.182461	0.025289
35	40	41.89441	3.588789	0.04736
36	41	39.70141	1.686336	0.031673
37	41	40.43241	0.322158	0.013844

The RMSE and MAPE for the validation data (days 31 and 37) are 1.5721 and 0.0332 (3.32%), respectively. Applying Eq. (13.40) that is, using the F_t values to forecast F_{t+1} instead of Y_t , we get the values in Table 13.23.

Table 13.23 | AR(1) model forecast using equation 13.40

Day	Y_t	F_t	$(Y_t - F_t)^2$	$ Y_t - F_t /Y_t$
31	45	44.0874	0.8328	0.0203
32	45	42.6893	5.3393	0.0513
33	46	41.6673	18.7723	0.0942
34	43	40.9202	4.3256	0.0484
35	40	40.3741	0.1399	0.0094
36	41	39.9749	1.0509	0.0250
37	41	39.6830	1.7344	0.0321

The RMSE and MAPE for the validation data (days 31 to 37) using Eq. (13.40) are 2.1446 and 0.04009 (4.009%), respectively.